

JOURNAL OF MEDICINE

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**EXPERIMENTAL AND CLINICAL GNOTOBIOTICS:
INFLUENCE OF THE MICROFLORA ON
GRAFT-VERSUS-HOST DISEASE AFTER
ALLOGENEIC BONE MARROW TRANSPLANTATION¹**

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Key Words: Bone marrow transplantation, gnotobiotics, graft-versus-host disease, human patients, mice.

Subjects: Human patients; mice.

Abbreviations: BMT = Bone marrow transplantation; GID = gastrointestinal decontamination; GvHD = graft-versus-host disease; GvHR = graft-versus-host reaction; HF = Houston flora.

Abstract

One of the major complications of allogeneic bone marrow transplantation (BMT) is graft-versus-host disease (GvHD), which

¹ Part of the experimental work described here was supported by the Netherlands Cancer Foundation grant RI 86-5.

is caused by donor type lymphocytes which react against the recipient's tissues.

An important factor which influences GvHD is the recipient's gastrointestinal microflora. This was originally observed in gnotobiotic mice. Infusion of 10^7 H-2 incompatible bone marrow cells into lethally irradiated (9.0 Gy X-rays) conventional mice results in a late onset type GvHD which causes the death of the majority of the recipients during the first two months after BMT. This mortality can be completely prevented if the recipients are germfree mice, or when they are conventional animals which have been subjected to complete or selective gastrointestinal decontamination (GID).

In a mouse model, the mechanism responsible for the influence of the microflora on GvHD after allogeneic BMT was investigated. These studies indicate that GvHD can be induced by activated T-lymphocytes from donor origin reacting against bacterial antigens which might be cross-reactive with the recipient's epithelial tissue antigens. Activation of these T-lymphocytes is confined to antigens of certain bacterial species of the recipient which are not present in the indigenous microflora of the donor mice. These bacteria most likely belong to the anaerobic flora of the recipient. The latter hypothesis is strongly supported by the observation in human patients that, in contrast to complete GID, selective decontamination of the gastrointestinal tract did not have any beneficial effect on moderately severe to severe GvHD after transplantation with MHC-matched sibling donor bone marrow grafts.

Introduction

Allogeneic bone marrow transplantation (BMT) is currently being used as treatment for many fatal diseases of the hemopoietic system, among them severe aplastic anemia (Löwenberg and Gale, 1985) and leukemia (Dicke *et al.*, 1985). Furthermore, patients suffering from fatal hereditary diseases that are associated with a dysfunction of the lymphoid system, like severe combined immunodeficiency (Dooren and Vossen, 1985) and patients with inherited severe metabolic disorders (Barranger, 1984) are being treated with bone marrow grafts. In total about 2500 patients are transplanted annually world-wide in 160 BMT centers.

One of the major complications of allogeneic BMT is graft-versus-host disease (GvHD), which is caused by donor type

lymphocytes which react against the recipient's tissues. According to an evaluation of data from 2036 recipients of HLA identical sibling bone marrow transplants reported to the International Bone Marrow Transplant Registry, moderate to severe GvHD occurred in about 45% of these patients. In 48% of them, GvHD was related to their death (Gale *et al.*, 1987). The severity of GvHD is influenced by several factors, which include the degree of immunogenic disparity (Uphoff and Law, 1958), the number of cells grafted (van Bekkum, 1964), the number of T-lymphocytes present in the graft (van Bekkum, 1964; van Bekkum, 1972), the donor's sex (Gale *et al.*, 1987) and the age of the recipient (Gale *et al.*, 1987).

Microflora And Graft-Versus-Host Disease In Mice

Another important factor influencing GvHD is the gastrointestinal microflora of the recipient. This was originally observed in gnotobiotic mice. Infusion of 10^7 H-2 incompatible bone marrow cells into lethally irradiated (9.0 Gy X-rays) conventional mice results in a late onset type GvHD which does not give rise to symptoms until about three weeks after BMT. This disease kills the majority of the recipients during the next two months but those that survive for more than three months seem to have recovered (van Bekkum and de Vries, 1967; van Bekkum *et al.*, 1974). Mortality attributable to this type of GvHD can be completely prevented if the recipients are germfree mice (Jones *et al.*, 1971; Truitt, 1978; van Bekkum *et al.*, 1974), or when they are conventional animals which have been subjected to complete (Heit *et al.*, 1973; Truitt, 1978) or selective gastrointestinal decontamination (van Bekkum *et al.*, 1974) by means of orally administered nonabsorbable antibiotics prior to transplantation. These findings suggest that not only histoincompatibility determines the occurrence and severity of GvHD, but that microflora-related factors also are of major importance. The hypothesis is that donor type lymphocytes can be stimulated or activated by antigens of bacteria from the gastrointestinal tract which cross-react with antigens present on the recipient's epithelial tissues, being the main targets in GvHD.

Clinical Bone Marrow Transplantation

In man, the effect of GID on GvHD has been controversial for a long time. Several studies reported a reduction of the incidence of acute GvHD after allogeneic BMT (Mahmoud *et al.*, 1984; Schmeiser *et al.*, 1984; Storb *et al.*, 1983). However, such an effect was not observed in other studies (Leblond *et al.*, 1987; Skinhøj

al., 1987; Storb and Thomas, 1985). Recently, we reported on a retrospective evaluation of the efficacy of GID in a protective environment for the prevention of GvHD in 65 children and adolescents, grafted consecutively for either severe bone marrow failure (n=29) or leukemia (n=36) (Vossen *et al.*, 1990). It was concluded that, in contrast to selective gastrointestinal decontamination (Group I, n=21), complete gastrointestinal decontamination (Group II, n=44) in a strict protective environment is a very effective method for preventing acute GvHD in children and adolescents; it resulted in a cumulative frequency of $\geq$ grade II acute GvHD of 17.5%, a low transplantation-related mortality of 26% and a good quality of survival in 69% of the graft recipients.

More recently the influence of complete and selective gastrointestinal decontamination in a strict protective environment on acute GvHD after allogeneic BMT was re-evaluated in a larger group of patients, which were transplanted in the Leiden Pediatric BMT center over a period of about 20 years. Since the above mentioned report, 37 more completely decontaminated children have been grafted (Group III). Major differences between the former study groups and group III were the use of methotrexate plus cyclosporin-A for GvHD prophylaxis in 25 graft recipients of group III, and the substitution of systemic (i.v.) antimicrobial prophylaxis for peroral GID early after BMT in recipients of group III with gastrointestinal complaints and bad compliance for oral antimicrobial drugs. Patients in all three groups received full bone marrow grafts from HLA genotypically identical siblings, following the usual pre-treatment regimens. The characteristics of the 94 evaluable patients are given in Table I.

All patients were maintained in a strict protective environment, i.e., in laminar down flow isolators (van der Waaij *et al.*, 1973), using aseptic nursing techniques and sterilization of food, beverages and all other items brought into the isolator (Vossen and van der Waaij, 1972).

Antimicrobial drugs and dosages administered orally for selective gastrointestinal decontamination in group I, complete gastrointestinal decontamination of patients in group II, and complete gastrointestinal decontamination in group III are given in Table II. Young children below the age of 2 years received half of the indicated dosages. The drugs were administered in four divided doses per day, except for co-trimoxazole, which was given twice daily. When during the early period after BMT children were unable to swallow the drugs for complete gastrointestinal decontamination

TABLE I

Characteristics of 94 Evaluable Patients

GID	Bone Marrow Failure			Hematological Malignancy	
	SAA	Fanconi	MDS	Leukemia	NHL
Selective (18)	9	2	1	6	0
Complete (76)	14	4	7	47	4

SAA = severe aplastic anemia; Fanconi = Fanconi's anemia; MDS = myelodysplastic syndrome; NHL = non-Hodgkin's lymphoma

TABLE II

Antimicrobial Drugs for Complete and Selective Gastrointestinal Decontamination (Daily Dose)

Group I	Group II	Group III
Nalidixic Acid (90 mg ^{a,b})	Neomycin (200 mg)	Gentamicin (800 mg)
Co-trimoxazole (12/60 mg ^{a,b})	Polymyxin B (2000 mg)	Cephalexin (2000 mg)
Neomycin (15 mg ^a)	Cephalexin (2000 mg)	Amphotericin B (2000 mg)
Polymyxin B (20 mg ^a)	Amphotericin B (2000 mg)	
Amphotericin B (2000 mg)		

^aper Kg body weight; ^bused only in a limited number of patients

due to nausea and vomiting, the suppression of the gut microflora was continued by i.v. administration of co-trimoxazole and cefamandole in usual therapeutical dosages; this was only done in children of group III. In individual cases of the same group, i.v. 5-flucytosine was added to the oral administration of amphotericin B, when elimination of yeasts from the gut had not been successful. This was done to suppress further growing of these microorganisms.

Decontamination started $\geq$ one week before the date of BMT and was given for at least a total of 40 days after BMT, according to experimental data (van Bekkum, 1977). Both complete and selective GID were considered successful when in the period from 7 days before until 40 days after BMT the target-microorganisms could not be isolated from more than two consecutive fecal samples. GvHD was diagnosed by clinical symptoms. The severity of acute GvHD was graded according to Thomas *et al.*, (1975).

According to the criteria used, GID was successful in 14 out of 18 evaluable children with selective GID in group I (78%); in 11 out of 40 evaluable children with complete GID in group II (27.5%) and in 19 of 36 evaluable children with complete GID in group III (53%).

The occurrence of $\geq$ grade II acute GvHD was 6/18 (33%) in group I, 7/40 (17%) in group II and 1/36 (3%) in group III. More relevant was the finding that 0/30 successfully completely decontaminated children developed $\geq$ grade II acute GvHD versus 6/14 (43%) successfully selectively decontaminated children. The latter is not different from the incidence (45%) in 2036 recipients of HLA identical sibling bone marrow transplants analyzed by the International Bone Marrow Transplant Registry (Gale *et al.*, 1987). From our observations it can be concluded that complete gastrointestinal decontamination is superior to selective gastrointestinal decontamination in preventing $\geq$ grade II acute GvHD.

The Underlying Mechanism

To study the mechanism, which underlies the influence of the gastrointestinal microflora on GvHD, H-2 different donor and recipient mice with a SPF and a conventional microflora were employed (Heidt, 1989). For this purpose a conventional microflora was imported from the M.D. Anderson Cancer Institute, Houston, TX, USA, called "Houston flora" (HF), since in our institute only

TABLE III

Composition of the Different Microfloras

SPF-Flora	Houston-Flora
Anaerobic microflora	Anaerobic microflora
Not Defined	Not Defined
Aerobic microflora	Aerobic microflora
<i>Streptococcus faecalis</i> (7173711 ^a)	<i>Streptococcus faecium</i> (7355510 ^a)
	<i>Streptococcus faecium</i> (7317550 ^a)
<i>Staphylococcus aureus</i> (6726153 ^b)	<i>Staphylococcus xylosus</i> (6736552 ^b)
<i>Staphylococcus epidermidis</i> (6706133 ^b)	<i>Staphylococcus haemolyticus</i> (6632171 ^b)
<i>Escherichia coli</i> (5144572 ^c)	<i>Escherichia coli</i> (5144572 ^c)
	<i>Escherichia coli</i> (5144532 ^c)
	<i>Proteus mirabilis</i> (0536000 ^c)
	<i>Proteus mirabilis</i> (0534000 ^c)
	<i>Pasteurella pneumotropica</i> (1220000 ^d)*

^aBiotype (API 20 Strep; API System, Montalieu-Vercieu, France);

^bBiotype (API Staph); ^cBiotype (API 20 E); ^dBiotype (API 20 NE);

* Isolated from nasal washings only.

SPF animals are being bred. The composition of the SPF flora and this HF is given in Table III.

Germfree C3H/Law breeding pairs were associated with the conventional flora. HF C57BL/Rij mice were obtained by foster nursing caesarian derived C57BL/Rij newborns by HF C3H/Law mothers. The HF C57BL/Rij animals obtained in this way were used as breeding animals to produce the experimental animals. Before entering the experiment, all mice were kept in Trexler type plastic film isolators to prevent undue association with any other microorganisms. Beside the above mentioned experiments using donor and recipient mice with the SPF flora or the HF, a second series of experiments was carried out to study the effect of complete and selective GID of HF recipients on GvHD. During both series of experiments, all recipients were housed under conditions of strict reverse isolation in a laminar cross flow isolator to prevent contamination of the animals with any new microorganisms (van der Waaij and Andreas, 1971). The animals received autoclaved (10 min., 134°C) AM-II food pellets (Hope Farms B.V., Woerden, The Netherlands) and acidified (pH 2.8) sterile drinking water.

The recipients were lethally (9 Gy) irradiated as a conditioning for BMT. The next day, they were injected i.v. with 10^7 bone marrow cells from C57BL/Rij donor mice. Irradiation of the mice and transplantation of the bone marrow cells were also performed under conditions of strict reverse isolation.

According to the microbiological status of the donors and the recipients, there were four different experimental groups in the first series of experiments. They were: HF recipients of SPF donor bone marrow (SPF *to* HF), HF recipients of HF donor bone marrow (HF *to* HF), SPF recipients of SPF donor bone marrow (SPF *to* SPF), and SPF recipients of HF donor bone marrow (HF *to* SPF).

No significant mortality from GvHD occurred in HF *to* HF recipients, SPF *to* SPF recipients and HF *to* SPF recipients, but the mortality of SPF *to* HF recipients was 80%. In the second series of experiments, prevention of GvHD (with a mortality-pattern identical to that in the HF *to* HF recipients, SP *to* SPF recipients and HF *to* SPF recipients) was observed in the completely decontaminated HF C3H/Law recipients of SPF C57BL/Rij bone marrow, while no effect of selective GID of the HF recipients was seen.

In the first series of experiments, lethal GvHD occurred in only one of the four transplanted groups of mice. This was the group of "conventional" (HF) recipients of SPF bone marrow (SPF

to HF), in which severe diarrhea was also observed and which showed the most distinct histopathological lesions. When the donors and recipients of the bone marrow were both HF carrying animals (HF to HF), more than 90% of the recipients survived after BMT. Also in the two other groups in which the recipients were SPF (i.e., SPF to SPF and HF to SPF), more than 90% of the recipients survived. This is comparable with observations made when recipients of allogeneic H-2 mismatched bone marrow are germfree or have been decontaminated (van Bekkum *et al.*, 1974). The results show that mortality due to GvHD after allogeneic H-2 mismatched BMT is significantly prevented when there is identity between the microflora of the donor and recipient (HF to HF and SPF to SPF) or when the recipients do not harbor any aerobic microorganisms other than those present in the donors (HF to SPF).

In previous publications it was hypothesized that certain bacteria belonging to the microflora of the recipient may play a role in the induction of delayed type GvHD after allogeneic BMT because they carry antigens which are cross-reactive with epithelial antigens. After transplantation of allogeneic bone marrow from a donor which was matched for the major histocompatibility complex (MHC), a mitigation of GvHD was observed in selectively decontaminated dogs (Vriesendorp *et al.*, 1981), which confirmed the findings in mice (van Bekkum *et al.*, 1974) suggesting that *Enterobacteriaceae* might play a role in the induction and severity of GvHD. The observation that selective GID of HF recipients did not mitigate GvHD, in contrast to complete GID of HF recipients does not support this hypothesis since *Enterobacteriaceae* are being eliminated by selective GID. Therefore, it has to be assumed that other microorganisms than *Enterobacteriaceae* are involved in the induction of GvHD. It is likely that bacteria belonging to the complex anaerobic microflora are responsible for the induction of GvHD. Foo and Lee (1974) found that antigenic cross-reaction exists between mouse intestine and a *Bacteroides* spp., one of the anaerobic members of the autochthonous microflora of the rodent gastrointestinal tract. The above mentioned conclusion is supported by the earlier mentioned observations that selectively decontaminated human recipients of allogeneic bone marrow developed significantly more $\geq$ grade II acute GvHD than completely decontaminated patients. The most striking difference between these two groups of patients was the presence of an anaerobic microflora in the first group of recipients (Vossen *et al.*, 1990).

The high mortality due to GvHD in HF recipients of SPF bone marrow in the first series of experiments can be explained by a double mechanism in evoking GvHD after allogeneic BMT. The first mechanism is the reaction of donor type T-lymphocytes against histocompatibility antigens. This graft-versus-host reaction (GvHR) gives rise to minimal intestinal lesions and subsequent limited mortality, as was the case in the completely decontaminated recipients. If however, the flora of the recipient of the allogeneic bone marrow contains anaerobic bacteria which do not belong to the indigenous flora of the donor, and which carry antigens cross-reactive with tissue antigens of the host, the donor T-lymphocytes are activated by these cross-reactive bacterial antigens and become reactive against host tissues. This results in a clinically and histologically more severe GvHR, leading to the death of the majority of the recipients.

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